

## **COORDINATION PARAMETERS**

### **1. PURPOSE**

According to IATA Worldwide Slot Guidelines (WSG), the coordination parameters are defined as the operational limits of all technical, operational and environmental factors at the airport. However, this definition is so broad that specific coordination parameters are not clearly understood.

It is therefore important that the various coordination parameters should be categorized and exemplified for better understanding by the coordinators and facilitators worldwide.

### **2. REFERENCES**

IATA Worldwide Slot Guidelines (WSG)

- Section 1.7 – What are the Key Principles of Airport Coordination?
- Section 4.5 – Role of the Facilitator
- Section 5.5 – Role of the Coordinator
- Section 6.1 – Demand and Capacity Analysis
- Section 7.1 – Key Principles of Schedule Facilitation
- Section 8.2 – General Priorities for Slot Allocation
- Section 9.3 – Availability of Airport Capacity
- Section 10 – Terms and Abbreviation

### **3. GUIDELINES**

The coordination parameters differ from airport to airport and from country to country. However, the coordination parameters can roughly be categorized into four factors, environmental factor, Air Traffic Control (ATC) factor, parking factor and terminal factor. Following tables are not exhaustive but gives the rough idea of coordination parameters at the airport.

When developing the coordination parameters at the airport, it should be carefully coordinated among the responsible authority, airport managing body, ATC service provider and aircraft operators. The coordination parameters should be provided to the coordinators and facilitators at least 7 days before the Initial Submission Deadline for each schedule season.

The coordination systems the coordinators and facilitators are using should have the functionality of manipulating those coordination parameters properly.

## 4. COORDINATION PARAMETERS

### 4.1. ENVIRONMENTAL LIMITS

| Factor               | Parameter         | Explanation                                                                                                                                                                                                                                                                                                                                                            | Example                                                                                            |
|----------------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Environmental Factor | Movement Limit    | A limit on the total number of movements in a specified period (eg, day, week, season or year) imposed for environmental reasons. The limit may apply during specified hours (eg. a night period).                                                                                                                                                                     | 460 movement/day, or 3,200 movements/week, or 170,000 movement/year or 7000 night movements/season |
|                      | Noise Quota       | A limit on the total number of 'noise points' in a specified period (eg, day, week, season or year) imposed for environmental reasons. The limit may apply during specified hours (eg. a night period).<br>'Noise Points' are assigned to each aircraft type, typically based on noise certification data with noisier aircraft types having more points per movement. | 26 points/night or 5200 points/season (during night period)                                        |
|                      | Night Curfew      | Certain hours of the night where aircraft operations are totally banned at the airport to protect local communities.<br>Note: where the curfew is defined by landing/takeoff times, the coordination parameters must include an appropriate taxi time as slots are allocated based on on/of block times.                                                               | Night curfew from 23:00 to 06:00 next morning                                                      |
|                      | Reduced Operation | Hours where capacity is limited for environmental reasons, typically during late at night and early in the morning at the airport to protect local communities.                                                                                                                                                                                                        | 28 movements/ 60min from 21:00 to 23:00<br>28 movements/ 60min from 6:00 to 7:00                   |

## 4.2. RUNWAY CAPACITY

| Factor                           | Parameter        | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Example |     |       |     |
|----------------------------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|-------|-----|
| ATC (Air Traffic Control) Factor | Hourly Movements | <p>The maximum number of aircraft movements in each 60 minute period (clock hours or rolling hours), typically expressed as a maximum number of Arrivals, Departures and Total movements.</p> <p>The arrival + departure limits may be higher than the total limit, providing some flexibility to switch arrival and departure slots (for a mixed-mode runway operation).</p> <p>The capacity may vary by hour-of-day for operational or environmental reasons, or be profiled to better match patterns of demand (eg, hours with mostly arrivals or mostly departures).</p> | Hours   | Arr | Dep   | Tot |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 05-18   | 48  | 50    | 68  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 19-22   | 36  | 36    | 48  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | or      |     |       |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Hour    | Arr | Dep   | Tot |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0600    | 15  | 35    | 45  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0700    | 25  | 31    | 50  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 0800    | 24  | 24    | 40  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | :       | :   | :     | :   |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2000    | 24  | 24    | 40  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2100    | 31  | 15    | 38  |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2200    | 27  | 20    | 36  |
|                                  | Sliding Scale    | <p>The possible combinations of arrival and departure movement limits are specified on a sliding scale in a tabular form, allowing flexibility between the mix of arrivals and departures in each time period.</p> <p>Typically total capacity is maximized when there is close to a 50/50 split between arrivals and departures; hours with mostly arrivals or departures will have lower total capacity.</p>                                                                                                                                                               | Arr     | Dep | Total |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2       | 30  | 32    |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | :       | :   | :     |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 16      | 18  | 34    |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 17      | 17  | 34    |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 18      | 15  | 33    |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | :       | :   | :     |     |
|                                  |                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 26      | 3   | 29    |     |

|         |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
|---------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|-----|-----|-----|-------|----|----|----|-------|---|---|---|----|--|--|--|-------|---|---|----|
|         | Sub-constraint | <p>In addition to hourly capacities, a narrower time interval sub-constraint of 5, 10, 15 or 30 minutes is used to smooth flights within the hour. In order to provide the best balance between scheduling flexibility for airlines and effective schedule smoothing, 10 or 15 min sub-constraints are useful.</p> <p>Typically the sum of sub-constraints across an hour is 10-20% higher than the hourly limit to provide some scheduling flexibility. For example, if the hourly limit is 40/hour, the sub-constraint might be 8-per-10 min or 12-per-15 min.</p> | <table><tr><td>Constr.</td><td>Arr</td><td>Dep</td><td>Tot</td></tr><tr><td>60min</td><td>24</td><td>24</td><td>40</td></tr><tr><td>10min</td><td>5</td><td>5</td><td>8</td></tr><tr><td>Or</td><td></td><td></td><td></td></tr><tr><td>15min</td><td>7</td><td>7</td><td>12</td></tr></table> | Constr. | Arr | Dep | Tot | 60min | 24 | 24 | 40 | 10min | 5 | 5 | 8 | Or |  |  |  | 15min | 7 | 7 | 12 |
| Constr. | Arr            | Dep                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Tot                                                                                                                                                                                                                                                                                            |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
| 60min   | 24             | 24                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 40                                                                                                                                                                                                                                                                                             |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
| 10min   | 5              | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 8                                                                                                                                                                                                                                                                                              |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
| Or      |                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
| 15min   | 7              | 7                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 12                                                                                                                                                                                                                                                                                             |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |
|         | Rolling Factor | <p>Coordination parameters may be fixed constraints, calculated at the same time interval of the constraint (eg, a 60min limit calculated every 60min), or rolling constraints calculated more frequently (eg, 60min limit calculated every 10 min). Rolling factors can help smooth the schedule, but are more complicated than fixed constraints.</p>                                                                                                                                                                                                              | <p>10 movements/ 15min with a rolling factor of 5min.</p>                                                                                                                                                                                                                                      |         |     |     |     |       |    |    |    |       |   |   |   |    |  |  |  |       |   |   |    |

#### 4.3. PARKING CAPACITY

| Factor                       | Parameter          | Explanation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Example                                                                                                                                          |
|------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Aircraft Type/<br>Stand Size | Stand Size         | <p>The size of aircraft which can be parked on a particular stand and/or the number of stands available by each size.</p> <p>The ICAO standard aircraft sizes are:<br/>           Code F – A380, B748<br/>           Code E – B747, B777, B787, A330, A340, A350<br/>           Code D – A300, A310, B757, B767, MD11<br/>           Code C – A320, B737, E170/190<br/>           Code A/B – GA/BA types</p> <p>Specific airports may have non-standard stand sizes, and specific stand categorization may be required</p> | APRON C    D    E    F<br>Alpha    6    1    9    3<br>Bravo    14    4    6    0<br>Charlie    20    0    9    0<br>Total    40    5    24    3 |
|                              | MARS stands        | MARS (Multiple Aircraft Ramp System) stands are stands that can either park, for example, 1 Code E or 2 Code C aircraft.                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                  |
|                              | Minimum Break Time | The time between the block out time of departing aircraft and the block in time of the arriving aircraft on the same parking stand. Typically it is set at 10 to 20 minutes (shorter for narrow-body aircraft; longer for wide-body aircraft).                                                                                                                                                                                                                                                                             | 15 minutes                                                                                                                                       |

#### 4.4. TERMINAL CAPACITY

| Factor         | Parameter           | Explanation                                                                                                                                                                                                                              | Example                                                        |
|----------------|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Passenger Flow | Terminal Allocation | Flights are allocated to terminals (eg, T1, T2, FRT, GA/BA) and sub-terminals (eg, domestic and international) based on allocation rules, typically based on parameters such as Airline, route, service type and/or flight number range. | T1 Domestic<br>Last/Next Country=<br>Domestic T1 International |

|  |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                |
|--|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
|  |                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Last/Next Country = all others                                                                 |
|  | Load Factors          | <p>Assumed LFs used to convert aircraft seats into passengers, typically based on historical data (eg, average LFs in busiest month of previous equivalent season).</p> <p>Different LFs may be used for different types of traffic (eg, domestic/shorthaul, longhaul, LCCs, Charter).</p> <p>Different LFs may be used for different days-of-week or periods of the season where there is significant variation.</p>                                                                   | <p>Dom = 70%</p> <p>L/haul = 85%</p> <p>LCC = 85%</p> <p>Charter = 95%</p>                     |
|  | Pax Flow Limits       | <p>Maximum number of passengers per time period (eg, 60 min). Passengers are calculated from aircraft seats and assumed LFs.</p> <p>Different time periods may be used to model different processes, depending on typical passenger reporting profiles (eg, 60 min for security or immigration, 2h or 3h limits for check-in).</p> <p>A sub-constraint (eg, 30 min limit) or rolling factor (eg, 60min rolling every 15min) may be used to prevent flight bunching within the hour.</p> | <p>T60 = 2000 pax</p> <p>T120 = 3600 pax</p> <p>T30 = 1200 pax</p>                             |
|  | Check-in counters     | Explicit calculation of number of desks required, based on check-in desk opening profiles (eg, for 100-150 seat aircraft, 2 desks from STD-180min to STD-30min), or from passenger reporting profiles and transaction times.                                                                                                                                                                                                                                                            |                                                                                                |
|  | Separation Constraint | As an alternative to Passenger Flow constraints, the minimum separation following the arrival or departure of a flight of a particular size (often used for small terminal facilities).                                                                                                                                                                                                                                                                                                 | <p>Up to 150 seats<br/>10min</p> <p>Up to 300 seats 20min</p> <p>More than 300 seats 30min</p> |